James Mynderse <jmynderse@ltu.edu>

Lawrence Tech Battery Pick & Place - SCHUNK Linear Actuators: Technical Info, Budgetary Info, Concept Model, and Trap Moves / RH

1 message

automation-apps.us.schunk.com <automation-apps@us.schunk.com> Fri, Dec 5, 2025 at 12:12 AM
To: "jmynderse@ltu.edu" <jmynderse@ltu.edu>
Cc: "Fischer, Douglas" <douglas.fischer@us.schunk.com>

Hello James,

I have reviewed the application data for the X-Z-Gripper Gantry for the Power Tool Battery setup.

I am recommending the SCHUNK Linear Actuators and Gripper (listed below) for the gantry system.

1. You will also find the budgetary information for the SCHUNK Linear Actuators and Gripper listed below.
2. I have provided the technical information each SCHUNK component that I am recommending for this project in this email.

I have provided a concept model of the SCHUNK components in the X-Z configuration.

1. The linear actuators in the model are setup to reflect the setups shown in the budgetary information.
2. I also added the KGG 100-80 Gripper in the gantry model.
3. The adapter plates that are shown in the gantry model are just for representation and they should not be used for design purposes.

I have also provided Trap Moves for each linear actuator to show the best possible move time, required motor torque, and total reflected inertia.

1. If we look at the each move time and the gripper open/close time (0.19/0.15 sec) then we would be looking at a total cycle time of 2.04 sec.
    a. Please note that this does not take into account dwell time that you might consider adding.
    b. If we need to go faster then it possible but we would need to look at a different gripper option as the acceleration from the actuators would increase the grip force required to hold the power tool battery.

We can discuss any questions that you may have in the meeting tomorrow.

SCHUNK Budgetary Information (X-Axis): Beta 100-ZRS Ball Screw Actuator at 550 mm Stroke

Qty (1)

List Price: $6,720.00 USD

Lead Time: 5-6 Weeks ARO (Note: Lead times are longer than normal due to the upcoming factory closures with the holidays – normal lead time is 3-4 Weeks ARO)

Beta 100-ZRS-40AT10-200-550-1050-AK-AZ2-2EO10-1ES10-6BL2-1-MGK2-GS24
Linear axis Beta
internal roller guide
toothed-belt drive
stroke per round: 200mm
repeatability accuracy(+/-): 0.08 mm
Mech. stroke: 550 mm
total length: 1050 mm
slide 1 short, 280mm
covered by cover band (AK)
with drive shaft (AZ2)
2 limit switch inductive, NC with 10m cable (EO10), Side D
1 limit switch inductive, NO with 10m cable (ES10), Side D
6 fastening strips size 2 (BL2)
motor flange (MGK)
coupling Typ 2.6

SCHUNK Budgetary Information (Z-Axis): Beta 80-SRS Ball Screw Actuator at 150 mm Stroke

Qty (1)

List Price: $4,870.00 USD

Lead Time: 5-6 Weeks ARO (Note: Lead times are longer than normal due to the upcoming factory closures with the holidays – normal lead time is 3-4 Weeks ARO)

Beta 80-SRS-M-2020-150-530-2EO10-1ES10-4NS3-1-MGK1-GS19
Linear axis Beta
internal roller guide
ball screw drive
pitch 20 mm
thread shaft diameter 20 mm
critical speed: 3000 rpm
with single nut (M)
Axial tolerance (standard): 0.08 mm,
repeatability accuracy(+/-): 0.03 mm
Mech. stroke: 150 mm
total length: 530 mm
slide 1 short, 210mm
covered by cover band (AK)
2 limit switch inductive, NC with 10m cable (EO10), Side C
1 limit switch inductive, NO with 10m cable (ES10), Side C
4 T-nuts M6 (4NS3)
motor flange (MGK)
coupling Typ 2.6

SCHUNK Budgetary Information: KGG 100-80 Parallel Gripper

Qty (1)

List Price: $1,210.00 USD

Lead Time: In Stock in North Carolina

ID #0303066

KGG 100-80

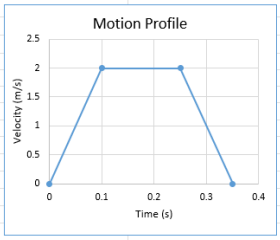
Gripper for small components pneumatic

- Stroke version 80

Simple Motion Sizer

Inputs

		Lawrence Tech: Beta 100-ZRS (X-Axis)
Payload (kg)	11	Battery (0.6 kg) + Gripper (1.1 kg) + Z-Axis (11.5 kg) + B100 Carriage (4.1 kg)
Total move distance (m)	0.5	Desired Move Distance = 500 mm (0.5 m)
Total move time (s)	0.35	Best Possible Move Time
Accel time (s)	0.1	
Additional external force (N)		
Overall lead (mm/rev)	40	200 mm/rev ÷ 5 (gear ratio) = 40 mm/rev
Efficiency (%)	90	
Friction coefficient	0.02	



Outputs

Max speed (m/s)	2.00
Max acceleration (m/s²)	20.00
Acceleration and friction force (N)	222.16
Max motor speed (rpm)	3000.00
Max motor torque (Nm)	1.57
Power (W)	493.68
Reflected Inertia (kg-m²)	0.00045

This motion sizer was designed as a quick reference tool. It does not take into account inertia values for drive components or the motor. It is up to the user to calculate the total lead of the system. For example, a 100mm/rev pulley driven by a 5:1 gear reducer is a 20mm overall lead. It is also up to the user to add in the weight of the payload in Newtons if the application is vertical.

Re-calc trap move

INPUTS:

Choose Actuator:	Beta
Belt Drive or Screw Drive	Belt Drive
Size	Beta 110-ZRS
Overal Length (mm)	1050
Coupling	GS24
Gear Ratio (Belt Drive Only)	5
Breakaway Torque (Nm)	0.7

OUTPUTS:

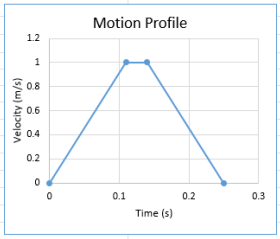
Required Motor Torque (Nm)	2.27
Total Reflected Inertia (kg-m²)	0.00117309

Note: The total reflected inertia (above) does not include the gearbox inertia.

Simple Motion Sizer

Inputs

		Lawrence Tech: Beta 80-SRS (Z-Axis)
Payload (kg)	11	Battery (0.6 kg) + Gripper (1.1 kg) + B80 Profile (9.3 kg)
Total move distance (m)	0.14	Desired Move Distance = 5.5" or 140 mm (0.14 m)
Total move time (s)	0.25	Best Possible Move Time
Accel time (s)	0.110000002	
Additional external force (N)	110	11 kg x 10 m/s² = 110 N
Overall lead (mm/rev)	20	Ball Screw Lead = 20 mm/rev
Efficiency (%)	90	
Friction coefficient	0.02	



Outputs

Max speed (m/s)	1.00
Max acceleration (m/s²)	9.09
Acceleration and friction force (N)	102.16
Max motor speed (rpm)	3000.00
Max motor torque (Nm)	0.75
Power (W)	235.73
Reflected Inertia (kg-m²)	0.00011

This motion sizer was designed as a quick reference tool. It does not take into account inertia values for drive components or the motor. It is up to the user to calculate the total lead of the system. For example, a 100mm/rev pulley driven by a 5:1 gear reducer is a 20mm overall lead. It is also up to the user to add in the weight of the payload in Newtons if the application is vertical.

Re-calc trap move

INPUTS:

Choose Actuator:	Beta
Belt Drive or Screw Drive	Screw Drive
Size	Beta 80-SRS
Overal Length (mm)	530
Coupling	GS19
Gear Ratio (Belt Drive Only)	
Breakaway Torque (Nm)	0.6

OUTPUTS:

Required Motor Torque (Nm)	1.35
Total Reflected Inertia (kg-m²)	0.00019697

Best Regards,

Rick Hunsucker

Team Lead Linear Actuators AC

SCHUNK

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4 attachments

Technical Information_SCHUNK KGG 100-80 Parallel Gripper.pdf
1921K

Technical Information_Beta 100-ZRS Belt Drive Actuator.pdf
628K

Technical Information_Beta 80-SRS-SSS Ball Screw Actuator.pdf
192K

Concept Model_Lawrence Tech X-Z Gantry_SCHUNK Components.zip
1474K